

Strategies of high-accuracy memristor-based analogue computing in memory for artificial intelligence

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Memristor-based analogue computing in memory (CIM) offers revolutionary gains in energy efficiency and computing power for data-intensive applications such as artificial intelligence. However, it typically struggles with achieving high accuracy at the same time, owing to the noise-sensitive nature of analogue computing and the non-ideal characteristics at the device and circuit levels that inevitably result in computing errors. Although progress has been made in device engineering and hardware–algorithm co-optimization to mitigate the error and parasitic effects, many of these advances inadvertently incur a hardware or energy consumption overhead, undermining the core benefits of analogue CIM. This Review dissects the computing error sources across the CIM hierarchy from the memristor device and array to the system architecture and algorithm, and evaluates the strategies to minimize those errors. We highlight the material and device innovations, array-level techniques and algorithm–architecture co-design frameworks towards high-accuracy analogue CIM. By dissecting the trade-off between computing accuracy and implementation cost, this Review draws a roadmap for translating memristor-based analogue CIM technology from proof-of-concept prototypes to large-scale deployment for accelerating next-generation artificial intelligence.

Artificial intelligence (AI) has progressed rapidly in recent years, and transformative technologies such as large language models (LLMs) now profoundly impact our lives¹. However, rising computational demands for AI substantially outpace the improvement in chip performance that is driven by Moore's law scaling, creating a critical hardware bottleneck². A primary reason for this challenge lies in the von Neumann architecture that underpins modern computing systems. This architecture necessitates frequent data shuttling between physically separated

memory and processing units (Fig. 1a), causing a fundamental von Neumann bottleneck alongside memory and power walls that critically constrain computing performance and energy efficiency³.

To overcome this fundamental constraint, memristor-based analogue computing in memory (CIM) offers a highly promising solution inspired by the biological brain⁴. It computes the matrix-vector multiplication (MVM)—the core operation of both artificial neural networks and signal processing—within the memory array following fundamental

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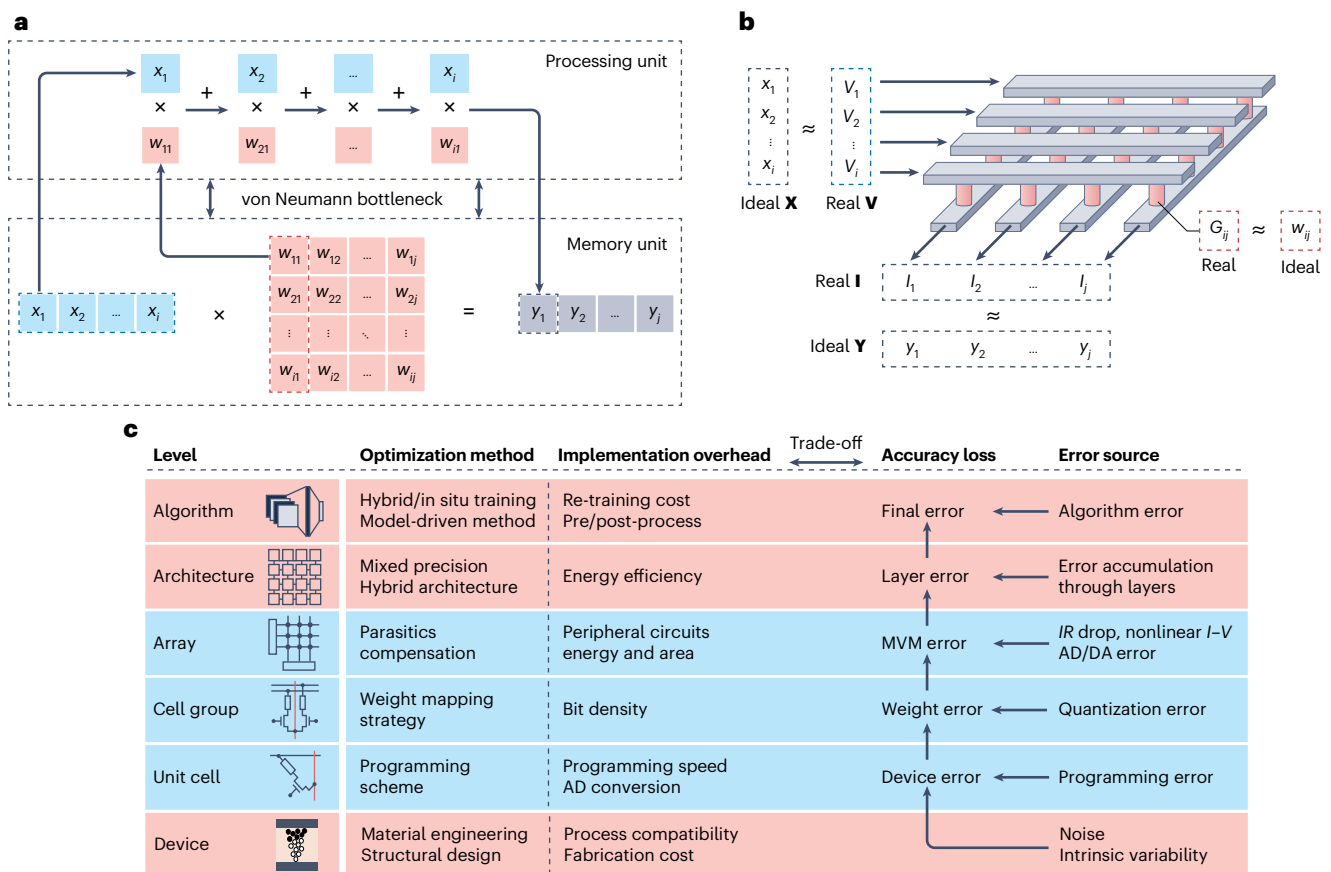


Fig. 1 | Memristor-based analogue CIM architecture for accelerating AI.

a, Digital computing schematic for MVM in the von Neumann architecture, depicting the von Neumann bottleneck that occurs between processing and memory units. **b**, Analogue CIM realized on a memristor crossbar array. The input vector X is mapped to input voltages V . Each element w_{ij} of the weight matrix W is mapped to a memristor conductance value G_{ij} . Through Ohm's law and Kirchhoff's current law, the resulting output current I is calculated, representing the output vector Y . Non-ideal factors introduce computing errors: input voltage

errors (for example, from digital-to-analogue (DA) conversion errors), variations in programmed conductance G_{ij} (for example, from device variability) leading to deviations between I and the ideal output Y . **c**, Parameters and optimization methods at different levels of the memristor-based analogue CIM hierarchy towards achieving high accuracy. Errors propagate and accumulate across hierarchical levels, whereas optimization strategies face a trade-off between implementation overhead and accuracy loss.

physical laws such as Ohm's law and Kirchhoff's current law (Fig. 1b). Performing computation directly where data resides delivers unparalleled advantages in energy efficiency and speed, effectively overcoming the constraints of the von Neumann paradigm. Substantial progress has been made in developing analogue CIM chips for AI, evolving from initial proof-of-concept arrays⁵ to board-level integrated systems^{6–9} and recently achieving full system-on-chip integration^{10,11}. Successful demonstrations across diverse domains including AI^{12,13}, signal processing^{14,15} and scientific computing¹⁶ underscore the potential of this technology to fundamentally revolutionize hardware for those data-intensive applications.

However, the inherent computing errors of analogue CIM, primarily from device-level fluctuations and array-level parasitics, limit its accuracy. Often overlooked in idealized theoretical analyses, these errors pose two challenges for practical hardware implementations. First, strategies to mitigate those errors usually introduce substantial overheads in hardware cost and energy consumption, which undermine the core advantages of analogue CIM. Second, these errors could accumulate from devices to arrays⁵ and hence fundamentally limit the scalability and accuracy of analogue CIM, restricting its practical applications for deep neural networks (DNNs) and scientific computing.

Therefore, addressing the error effect and improving the accuracy with acceptable overhead is a pivotal challenge for translating memristor-based analogue CIM from laboratory prototypes to

large-scale adoption. Whereas previous research studies and reviews^{4,17,18} have investigated the device innovations, crossbar architectures or hardware–algorithm co-optimization strategies, they rarely scrutinize the accuracy–overhead trade-off that is inherent to these solutions. To fill this gap, this Review systematically analyses the error sources across the CIM hierarchy, from memristor device and array to system architecture and algorithm, and further evaluates the strategies for mitigating these computing errors, as shown in Fig. 1c. Through a comprehensive review, we intend to identify the useful approaches that improve the accuracy with acceptable overhead, providing a roadmap to bridge the gap between theoretical promise and real-world deployment.

Memristor design and device optimization strategies

As the fundamental cornerstone of analogue CIM, the key device properties of a memristor primarily include analogue switching characteristics and reliability metrics⁴. These analogue switching characteristics, which include the forming voltage, operation voltage, on/off ratio, linearity and symmetry, form the performance foundation for analogue CIM systems¹⁹. Meanwhile, the reliability metrics, including conductance drift or relaxation, device-to-device (D2D) variability, endurance and yield, also represent critical sources of CIM errors²⁰. To overcome these limitations, recent advances have mainly focused on

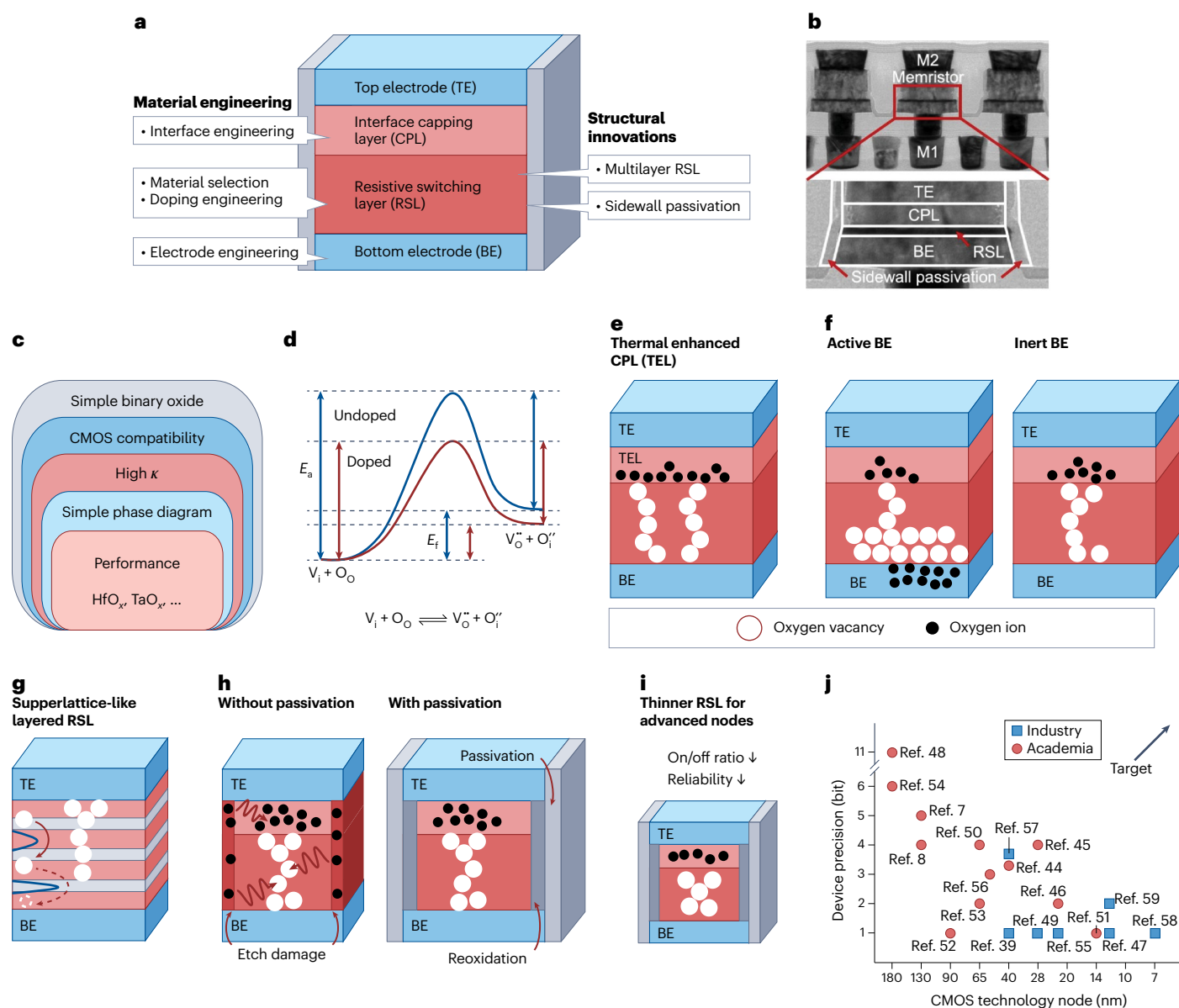


Fig. 2 | Memristor optimization strategies and CMOS integration challenges for analogue CIM. **a**, Overview of memristor device optimization approaches. **b**, Cross-sectional transmission electron microscopy image of a typical memristor integrated between metal 1 (M1) and metal 2 (M2) in 28 nm Si CMOS technology. **c–f**, Material engineering strategies, showing the material selection guidelines for the RSL (**c**), doping engineering to achieve a lower forming voltage (**d**), thermal enhanced layers (TELs) that enable multiple weak conductive filaments (CFs) for analogue switching characteristics (**e**) and the impact of active (left) and inert (right) bottom electrodes (BEs) on the morphology of the

CFs (**f**). E_a , activation energy; E_f , formation energy. White circles denote oxygen vacancies; black circles represent oxygen ions (as also in **g–i**). **g, h**, Structural innovation strategies, where a superlattice-like layered RSL segments the CFs into discrete domains separated by interface barriers (**g**) and the damage caused by dry etching (left) is mitigated via sidewall passivation (right) (**h**). **i, j**, CMOS integration challenges, where a thinner RSL benefits advanced-node integration but degrades the on/off ratio and reliability (**i**) and a degradation trend exists between device bit precision and CMOS technology node scaling (**j**)^{7,8,39,44–59}.

material engineering and structural innovations (Fig. 2a). This section first examines optimizations realized through material engineering applied to the resistive switching layer (RSL) and relevant interfaces. Next, it explores the structural innovations aimed at improving performance and reliability. Finally, the large-scale integration of optimized devices with advanced Si complementary metal–oxide–semiconductor (CMOS) processes (Fig. 2b) is discussed. Subsequent paragraphs detail these strategies.

Material engineering

Central to device optimization, material engineering focuses substantially on optimizing the RSL. Selecting suitable materials determines

foundational properties of the RSL. Simultaneously, doping engineering enables the fine tuning of device characteristics, while interface engineering and electrode engineering achieve targeted modifications of the upper and lower interfaces, respectively. These intertwined strategies collectively underpin the device development of memristor technology. Subsequent subsections detail the design principles underlying each method.

Material selection. The physical realization and experimental demonstration of the earliest memristors were reported using TiO_x -based devices^{21,22}. Subsequent extensive research has been conducted on a variety of oxides as the RSL, and most of them have focused on simple

binary oxides for convenient processing. Reliable resistive switching requires thermal stability of both the conductive and insulating phases, which is more readily achieved in materials with fewer thermodynamically stable solid-state phases²³. In addition, the thermo-chemical model predicts that a material's breakdown strength is inversely proportional to the square root of its dielectric constant (κ), thus favouring high- κ oxides to achieve low forming voltages²⁴. Compatibility with CMOS technology is also a critical consideration for advanced integration. Consequently, HfO_x and TaO_x have emerged as highly promising RSL materials that meet these criteria²³ (Fig. 2c). Memristors with high-entropy oxides²⁵ (for example, comprising Zr, Hf, Nb, Ta, Mo and W) exhibit a 'cocktail effect' that combines the benefits of the good retention of $\text{HfO}_x/\text{TaO}_x$ -based memristors and the gradual conductance modulation of WO_x -based memristors, enhancing the overall performance through the synergy of composite oxides.

Beyond amorphous oxides, other materials have been gaining interest as the RSL. For example, epiRAM (epitaxial random access memory) devices use epitaxial single-crystal SiGe films with predefined dislocations to guide Ag filaments, improving uniformity and enabling analogue switching²⁶. Brownmillerite oxides such as $\text{SrCoO}_{2.5}$ leverage ordered oxygen vacancy channels for controllable phase transitions²⁷. Molecular memristors have also demonstrated 14-bit programmable conductance states²⁸, although their practical maturity requires further validation given the limited independent replication studies so far²⁹. Overall, memristors based on those materials remain at an early stage of development, necessitating extensive further research for CMOS integration³⁰.

Doping engineering. Doping engineering, using metal cations (for example, Al, Ti, Ge, Mg), non-metallic ions (for example, Si, N, F) or ionic groups (for example, NO_2^-), provides a versatile strategy for modifying memristor characteristics. A primary goal is to reduce the forming voltage by lowering the formation energy of the Frenkel defect pair³¹ (Fig. 2d). In addition, dopants can introduce new conductive pathways or additional electrochemical processes to achieve voltage reduction³². Another key objective involves stabilizing oxygen vacancies at fixed positions, creating a 'pinning effect'. This promotes the formation of multiple, spatially confined weak filaments, substantially enhancing switching uniformity and device reliability during operation³³.

Interface engineering. Interface capping layers enhance the quality of the top interface and the device performance. For instance, Ti-based metallic interface capping layers act as effective oxygen reservoirs to substantially reduce the operation voltages³⁴. In addition, thermal enhanced layers (TELs) with a low thermal conductivity (for example, TaO_x), concentrate Joule heating in the HfO_x RSL, promoting the formation of multiple weak CFs and analogue switching³⁵ (Fig. 2e). Recent work has further proposed ternary oxide TELs with elements of divergent oxygen affinities: high-affinity elements (for example, Ti) form metal oxides to tune the conductance, whereas low-affinity elements (for example, Ta) remain metallic to bond oxygen ions, suppressing relaxation³⁶.

Electrode engineering. The material of the bottom electrode critically influences the memristor characteristics through interactions at the RSL/bottom electrode interface (Fig. 2f). Active electrodes, such as Ti-rich TiN, readily scavenge oxygen ions from the RSL, generating vacancies that lower forming voltage but accelerate endurance degradation via depletion of the oxygen reservoir³⁷. Conversely, inert electrodes (for example, Ru) mitigate such reactions due to high oxygen diffusion and reaction energy barriers, greatly enhancing the endurance³⁸. Alternatively, incorporating a weaker oxygen reservoir layer above the bottom electrode could also improve the device retention by anchoring the CFs³⁹.

Structural innovations

Beyond material engineering, structural innovations, such as multi-layer RSL and sidewall passivation, offer complementary routes to synergistically enhance the memristor performance.

Multilayer RSL. Multilayer RSL configurations aim to improve the performance through structural design. For instance, inserting an oxygen diffusion barrier such as SnO_2 in the HfO_x -based RSL confines filament rupture during reset, enhancing high-resistance state uniformity⁴⁰. Superlattice structures (for example, $\text{HfO}_x/\text{AlO}_x$ stacks) segment the filament into domains (Fig. 2g), forcing stepwise growth/rupture that enables gradual analogue switching and reduces variability⁴¹. Such a multilayer RSL can also elevate the crystallization temperature, mitigating thermal diffusion and grain-boundary-assisted ion migration to enhance retention⁴².

Sidewall passivation. Nanoscale memristors are vulnerable to etching-induced damage and free-oxygen diffusion near the cell peripheries, which destabilize the CFs while impairing reliability and uniformity³⁴ (Fig. 2h). Sidewall passivation addresses these scaling challenges by mitigating environmental interference at the cell edges. These strategies typically involve low-damage etching to reduce edge defects, cell-edge oxidation to form inert zones that confine switching, and encapsulation with oxygen barriers such as Si_3N_4 to seal the cell and stabilize the filament environment⁴³. This integrated passivation approach is essential to achieving reliable nanoscale memristors⁴⁴ and high yields⁴⁵ in large-scale integration.

Large-scale integration with advanced Si CMOS technology

Whereas the above optimizations improve the intrinsic device properties, the ultimate benchmark lies in the performance of large-scale arrays. Integration with Si CMOS technology enhances the yield and reduces variability, whereas scaling promises a greater density and efficiency for analogue computing. However, at advanced nodes, this integration introduces system-level challenges that require the co-optimization of device precision and low-conductance states.

Device precision. Achieving high-precision conductance modulation is a fundamental requirement for accurate analogue computing. However, a trade-off emerges between the technology node advancement and the achievable bit precision of memristors^{7,8,39,44–59}, as illustrated in Fig. 2i,j. This challenge stems partly from increased forming voltages for scaled memristor devices, which impose a critical challenge for the short-channel Si access transistor in advanced CMOS technology^{32,34}. Reducing the RSL thickness effectively lowers the forming voltage, yet paradoxically degrades critical device properties. A thinner RSL hinders complete CF rupture during reset operations, reducing the on/off ratio. Concurrently, shortened CFs exhibit increased sensitivity to ionic and thermal perturbations, compromising the device reliability^{60,61}. These factors collectively constrain the achievable device bit precision, making the development of high-precision memristors that are compatible with advanced nodes a critical challenge.

Low-conductance engineering. Beyond the intrinsic device precision, engineering the memristor to achieve a lower operational conductance state represents a crucial yet often overlooked strategy for mitigating array-level non-idealities. Although a high conductance usually benefits the read speed in memory applications, a lower conductance is paramount for CIM systems. It directly mitigates the voltage drop (IR drop, where I is the current and R is the resistance) along metal lines, reduces the total power consumption and prevents saturation in peripheral circuits, thereby enhancing the scalability and accuracy of large-scale CIM arrays. Strategies to lower conductance include reducing the cell area and increasing the RSL thickness, although these may elevate the forming voltage⁶². Alternatively, adjusting the oxygen

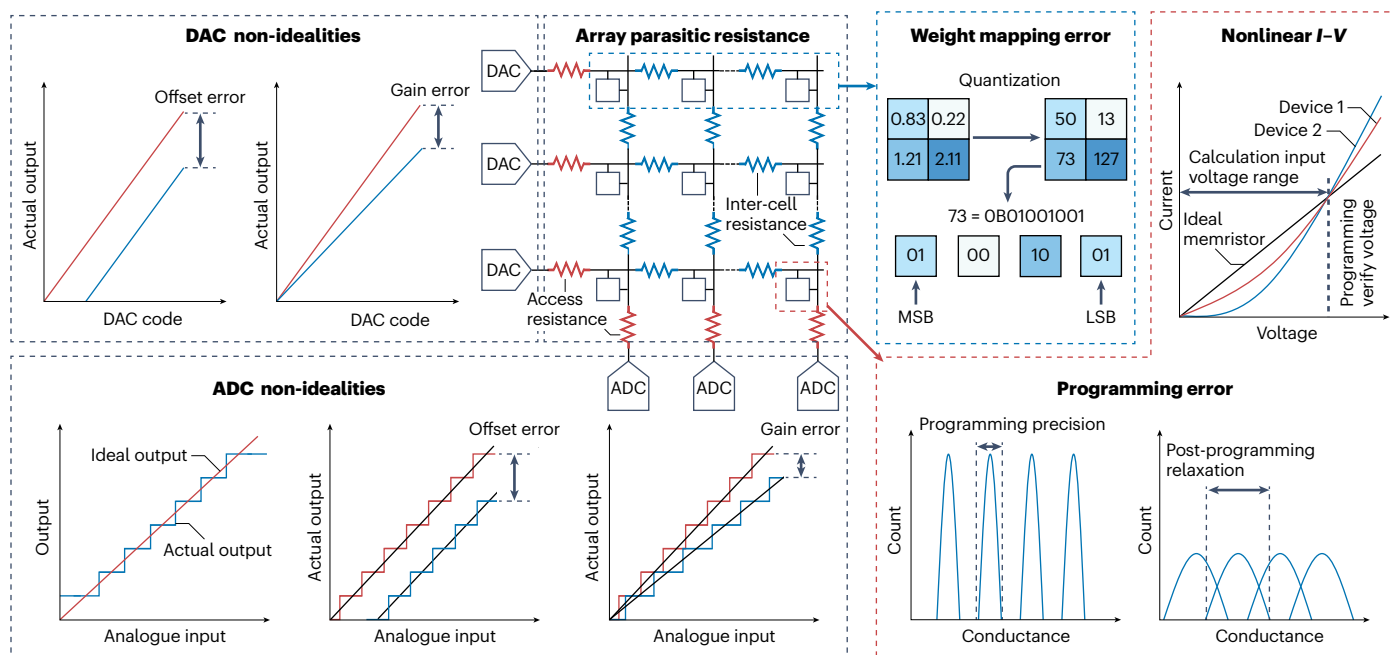


Fig. 3 | Various error sources from non-idealities in analogue CIM arrays.

Computational errors stem from multiple sources across hierarchy levels: unit cell non-idealities (programming-precision limitations, post-programming relaxation and nonlinear I - V characteristics); weight mapping quantization

error; array interconnect IR drops; and peripheral circuit non-idealities (DAC/ADC offset/gain errors, ADC quantization/swing limits). MSB, most significant bit; LSB, least significant bit.

content to increase the resistance of the TEL, thereby introducing a larger internal series resistance, offers an effective pathway to reduce the overall device conductance³⁶. However, it shall be noted that the memristor conductance cannot be arbitrarily lowered, as excessively low values may challenge reliable sensing and potentially degrade effective device precision⁶³. In summary, the co-optimization of low conductance with other device metrics presents a unifying direction for overcoming multiple device- and system-level limitations in analogue CIM.

Array-level strategies for error suppression

Beyond individual device optimization, the assembly of memristors into crossbar arrays for parallel MVM computations introduces additional error sources. Ideally, passive memristor crossbars, utilizing a single memristor (1R)^{5,64} as a unit cell, offer the highest cell density and three-dimensional (3D) stacking potential. However, the scalability of large-scale 1R arrays is constrained by intrinsic issues such as device variability⁶⁵ and sneak paths⁶⁶. These sneak paths refer to parasitic currents along unintended pathways in the array, which can degrade the write and read accuracy of memristor cells. Although self-rectifying memristors, which exhibit minimal current under a small read voltage, offer a promising solution⁶⁷, their efficacy demands further large-scale validation. As the most common solution, integrating an access transistor with each memristor in a one-transistor-one-memristor (1T1R)⁶⁸ configuration effectively suppresses sneak paths while also enabling precise conductance programming through stable current control⁶⁹. For signed weights, differential structures such as 2T2R⁷⁰ offer viable solutions. Various alternative unit cell designs, including 3T2R⁴⁴ and 2T1R⁷¹ configurations, have also been proposed to achieve specialized functionalities. However, incorporating transistors results in substantial area overhead and also limits the 3D stacking potential¹⁷. The 1TnR scheme⁷² aims to improve the cell density but introduces critical challenges for efficient array programming due to the sneak-paths issue. Integrating passive selector devices with memristors (1S1R)⁷³ is an active research area, yet the performance limitations of selectors

compared with well-established transistors restrict their practical utility in large arrays.

In addition, achieving sufficient computational accuracy in practice requires the inherent precision limitations of unit cells to be addressed^{74,75}. For example, multiple unit cells can be grouped to represent a single weight value with higher precision. In this case, peripheral circuits, including digital-to-analogue converters (DACs) and analogue-to-digital converters (ADCs), are also essential components and need to be taken into consideration⁷⁰. However, these array-level integrated systems are subject to numerous non-ideal sources⁸ that contribute to computational errors hierarchically, as shown in Fig. 3. At the unit cell level, programming of the memristor inevitably introduces errors due to the limited precision of the programming circuitry, where a certain margin within the target conductance value is predefined in the write-verify process. More importantly, the post-programming relaxation induces a conductance drift that exacerbates errors over time. Furthermore, the mapping of high-precision neural network weights (for example, single-precision floating-point format) onto groups of memristor devices also inherently introduces quantization errors. Meanwhile, array-scale parasitic resistances, including the inter-cell and access resistance, generate an IR drop during the MVM computation, and this effect is worsened as the array dimension is increased (for example, $>128 \times 128$)¹⁵. Peripheral circuitry introduces further non-idealities. For example, DACs and ADCs exhibit intrinsic offset and gain errors, and ADCs also suffer from quantization errors and input swing limitations. Furthermore, the nonlinear current-voltage (I - V) characteristics of memristors, although a device-level non-ideality, manifest their adverse effects specifically during array-based computation. Even under precise programming, such nonlinearity causes the output current to deviate from the ideal value predicted by Ohm's law, thereby introducing errors into the weighted-sum operations. By contrast, the device programming nonlinearity and D2D variability, which are critical for on-chip learning, usually do not introduce analogous errors for inference, as the conductance can be iteratively programmed to the target value.

Several studies have investigated the impact of various error sources on inference accuracy^{76,77}. On the basis on those analyses, Supplementary Table 1 provides an illustrative example of the hardware requirements to achieve iso-accuracy across different neural networks. However, the relative impact of these error sources on the computational accuracy largely depends on the application and system configuration. In some demonstrations, the initial computation accuracy can approach the software baseline but degrades over time, indicating that the post-programming relaxation effect is the dominant error source^{78,79}. Furthermore, the sensitivity to specific non-idealities, such as the programming error, *IR* drop and ADC quantization error, can differ by orders of magnitude depending on the specific neural network task⁷⁷. Even within a single network, different layers can exhibit distinct tolerances. For instance, in a recurrent neural network transducer, the fully connected layer is more sensitive to weight quantization than the long short-term memory (LSTM) layer¹³. To systematically address these challenges, the subsequent sections detail the various strategies that have been devised to mitigate these errors across different levels, outlining their specific design implementations and associated costs.

Device programming schemes

Accurate analogue CIM necessitates multi-bit-precision devices, which critically depend on reliable programming schemes to achieve the target conductance states. This choice of programming strategy profoundly affects both programming errors, including accuracy and post-programming conductance stability, and implementation costs. These costs include programming duration, power consumption and additional circuit overhead. Therefore, optimizing programming strategies becomes essential for balancing computational precision with practical constraints in hardware development.

Programming precision. Enhancing the programming precision has been the focus of extensive research on individual memristor devices and arrays. Early approaches relied on current-limitation techniques as in Fig. 4a, which included setting the compliance current via access transistor gate control⁸⁰ or external circuits⁸¹. These methods offer high parallelism but rely heavily on the device uniformity, typically limiting the achievable precision to 1 bit and rarely exceeding 3 bits. To improve the precision, fixed-condition write–verify schemes⁸² were introduced (Fig. 4b). Here, the programmed device conductance is read and compared with the target value after each write pulse to determine if subsequent set or reset operations are needed. While effectively improving the programming precision, it substantially increases circuit complexity and time cost due to the time-consuming verify steps, reducing the programming parallelism and demanding high device programming linearity. To address the linearity issue, incremental write–verify schemes⁸³—also known as incremental step pulse programming (ISPP), as commonly used in NAND flash memories—were adopted as illustrated in Fig. 4c. Previous studies have indicated that incremental steps in write voltage pulse amplitude, particularly via the word line^{84,85}, offer superior efficiency compared with incremental steps in pulse width⁸⁶. However, it must be pointed out that the incremental schemes, although further enhancing precision, require more complex control logic and additional caching of write conditions, further limiting the programming parallelism. The reported programming precision remains below 8 bits⁸⁷, limited by random telegraph noise from incomplete CFs near the main conduction channel^{88,89}. Applying small, alternating positive and negative denoising pulses can annihilate these channels, drastically reducing fluctuation and enabling 11-bit precision⁴⁸. This method requires repeated read/write steps, incurring time and energy overhead. However, the small voltages and parallel column-wise operation can mitigate these costs, making it suitable for inference applications with infrequent programming. For scenarios that demand frequent weight updates, such as on-chip

learning, the peripheral power budget in integrated large-scale arrays requires further evaluation.

To compare different programming schemes, here we define an efficiency figure of merit (FOM), which is the reciprocal of the product of hardware overhead and programming time, as elaborated in Supplementary Note 1. Figure 4d presents the resulting FOM values based on this analytical framework and our circuit-level estimations, illustrating the corresponding theoretical trade-offs. Whereas the actual conditions may vary, a distinct trend is observed between the estimated programming precision and the efficiency FOM across the evaluated schemes. Hybrid techniques, such as combining coarse adjustment using current limitation with fine tuning via incremental schemes, offer a compromise^{48,90}. This persistent challenge underscores the demand for more innovative device programming schemes that jointly optimize precision and overhead.

Post-programming relaxation. Beyond programming-precision limitation, a measurable post-programming relaxation effect is commonly observed in memristors, characterized by an unpredictable conductance drift after devices are programmed to a target state³⁶. This drift severely compromises the actual device precision, often rendering it considerably lower than the initially programmed precision⁷⁹. Repeated programming⁸ offers a straightforward and effective countermeasure, as illustrated in Fig. 4e. However, this approach is time-consuming and faces scalability challenges in large-scale arrays. Research on waveform engineering has identified specific waveform shapes, such as triangular⁹¹ or positive–negative pulse pairs⁹² (Fig. 4f), that can mitigate the relaxation effect to a certain extent. Furthermore, introducing increased waiting interval (T_{delay}) values between write and verify steps in the write–verify schemes has been found to reduce relaxation^{78,93} (Fig. 4g). Investigations have also revealed that the specific sequence of set and reset operations influences the relaxation behaviour^{79,91,94} (Fig. 4h). Additional studies have proposed reducing the voltage change between programming steps (V_{step})⁹⁵ (Fig. 4i), particularly for minimizing the conductance change of the final write operation (ΔG_{last})⁷⁹, as an effective drift countermeasure (Fig. 4j). Some studies have attempted to reduce the conductance state overlap through non-uniform state distributions⁹⁶, which is advantageous for memory applications. However, such schemes are less suitable for analogue CIM, as they require substantial algorithmic adaptations to account for nonlinear quantization. Recent strategies have increasingly combined multiple relaxation-mitigation techniques and prioritized overhead reduction⁷⁹. Nevertheless, most methods inevitably incur additional costs in hardware design complexity, programming duration and caching requirements, as summarized in Supplementary Table 2. Consequently, the development of more efficient, relaxation-aware programming schemes remains a critical objective.

In addition, optimizing the programming scheme of forming, a critical one-time process that determines CF formation, offers an efficient strategy to minimize relaxation. Approaches such as high-temperature forming⁶¹ and reverse forming⁹⁷ have shown promise in reducing relaxation by altering the filament morphology and modifying the distribution of oxygen ions and vacancies. Refining these forming strategies continues to be a key focus for future research.

Weight mapping techniques

How to map the synaptic weights of neural networks into the memristor device conductances is another key factor that determines the accuracy of analogue CIM. Whereas single devices demonstrating up to 2,048 and 16,520 conductance states have been reported^{28,48}, practical analogue CIM systems typically utilize only 2–32 states^{7–9,46}. This limited range proves to be insufficient for demanding computations, and introduces unavoidable quantization errors when converting high-precision weights. Consequently, efficiently mapping high-precision weights using groups of memristors has emerged as an

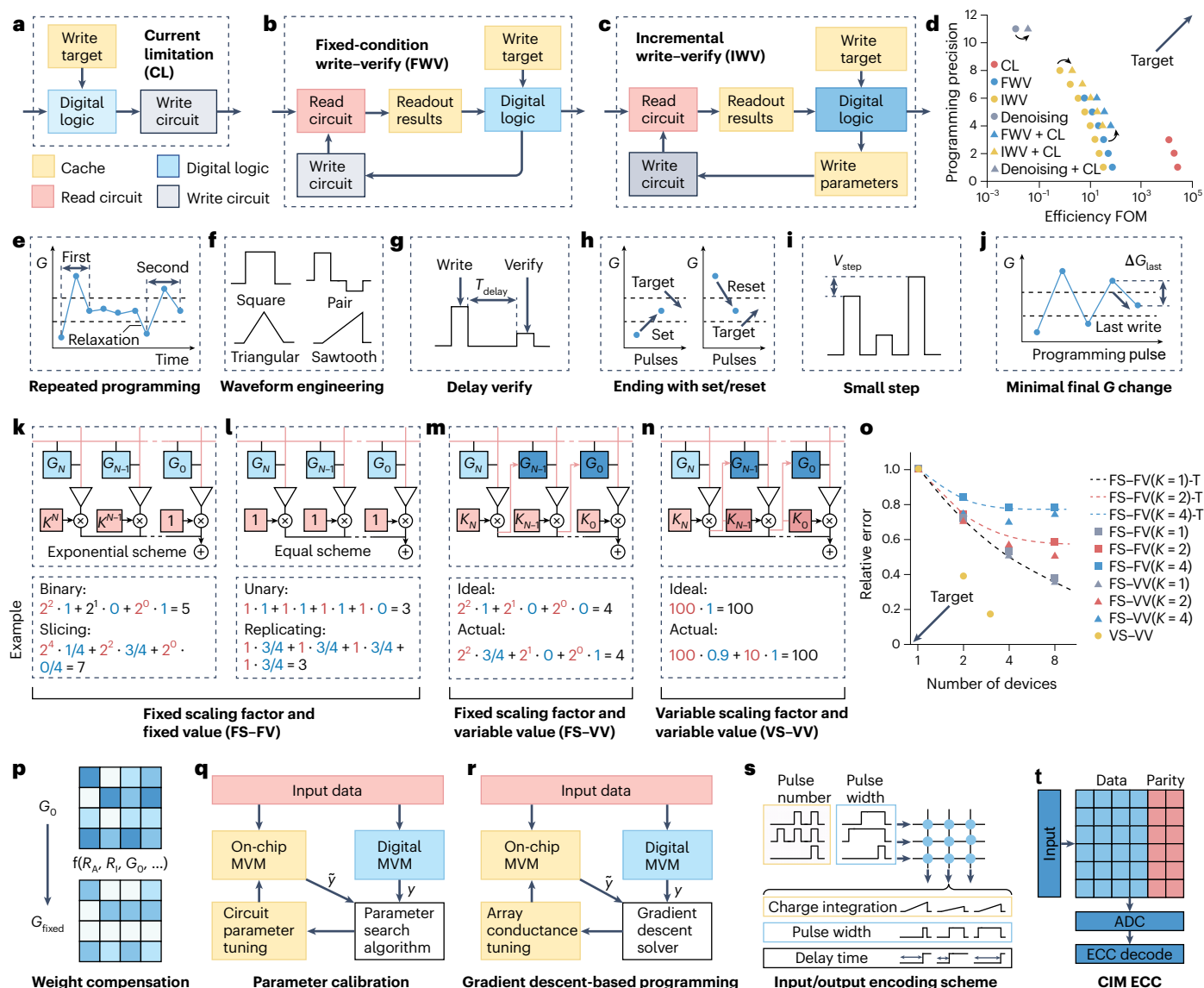


Fig. 4 | Array-level techniques for the enhancement of analogue CIM accuracy.

a–c, Programming scheme for high bit precision in memristors, including current limitation (**a**), fixed-condition write-verify (**b**) and incremental write-verify (**c**) methods. The colours of the blue boxes that represent ‘digital logic’ and the grey boxes that represent ‘write circuit’ are darkened to denote increasing complexity. **d**, On the basis of the methods used in **a–c**, plot showing the trade-off between the programming precision and the efficiency FOM, analysed in Supplementary Note 1. **e–j**, Strategies for mitigating post-programming relaxation include methods of repeated programming (**e**), waveform engineering (for example, triangular, positive–negative pulse pairs) (**f**), delay verify (**g**), ending with set/reset (**h**), small-step programming (**i**) and minimizing the final write conductance change (**j**). **k–n**, Mapping strategies include schemes comprising a fixed scaling factor and fixed value (FS–FV) with an exponential scaling factor (**k**), FS–FV with an equal scaling factor (**l**), a fixed scaling factor and variable value (FS–VV) (**m**) and a variable scaling factor

and variable value (VS–VV) (**n**). The lighter boxes and numbers denote fixed values/scaling factors, and the darker boxes denote values/scaling factors to compensate for the residual error. In **k**, K^N, K^{N-1} and K^0 denote exponential scaling factors with common base K ; in **m** and **n**, K_N, K_{N-1} and K_0 denote the scaling factors assigned to different devices. **o**, From the schemes in **k–n**, plot showing the relative error (the standard deviation ratio of the N -device combined weight with a single device) versus the number of devices. Theoretical curves (denoted by -T) and experimental data are shown, and K denotes the exponent base (where $K = 1$ means an equal scaling factor). **p–t**, Peripheral circuit precision compensation strategies include pre-distortion weight compensation for IR drop (R_A , access resistance; R_i , inter-cell resistance) (**p**), on-chip search-based parameter calibration (where y is the ideal output and $\tilde{y}$ is the actual output) (**q**), iterative refinement of conductance based on gradient descent (**r**), novel input/output encoding schemes (**s**) and error correction code (ECC) schemes tailored for CIM (**t**).

important research direction. Inspiration from digital systems has led to schemes that assign $N + 1$ memristors with fixed conductance values ($G_N, G_{N-1}, \dots, G_0$) and predefined exponential scaling factors ($K^N, K^{N-1}, \dots, K^0$), yielding a mapped weight of $W = G_N \times K^N + G_{N-1} \times K^{N-1} + \dots + G_0 \times K^0$ (Fig. 4k)⁹⁸. However, the dominant error contribution from higher-scaling-factor devices limits the benefits of adding more low-scaling-factor devices for precision enhancement. Both theoretical and experimental studies⁹⁹ have indicated that assigning memristor devices with equal scaling factors minimizes the overall

error, including unary¹⁰⁰ and replicated⁷⁴ forms (Fig. 4l). Assuming that the programming errors follow Gaussian distributions for identically programmed devices, the error for N devices averages as $1/\sqrt{N}$ of a single device’s error⁹⁹. The above solutions can be summarized as fixed scaling factor and fixed value (FS–FV) schemes. To overcome this theoretical limit, fixed scaling factor and variable value (FS–VV) schemes have been proposed⁹⁸ (Fig. 4m). These schemes program high-scaling-factor devices first and then use low-scaling-factor devices to compensate for the residual error, reportedly achieving a lower

error than equal-scaling-factor schemes using the same number of devices. Further refinement leads to variable scaling factor and variable value (VS–VV) schemes⁷⁵ (Fig. 4n), which dynamically optimize both the scaling factor assigned to each device and the device conductance (value) to compensate for the residual errors.

Here we evaluate the relative error versus the number of devices for these three techniques on the basis of the reported data^{75,99}, as shown in Fig. 4o. FS–FV schemes align with theoretical expectations, confirming that the equal-scaling-factor scheme outperforms the exponential scheme. FS–VV schemes show a slight advantage over FS–FV, whereas the VS–VV scheme achieves the lowest relative error. It is noteworthy that the implementation of such compensation schemes usually involves inherent trade-offs. The need to dynamically determine scaling factors and compensation values introduces computational overhead, which can be potentially mitigated by integrating the calculations into the write–verify process and using dedicated hardware. More importantly, supporting variable scaling factors inherently introduces area overhead. Mitigating this cost is likely to require strategies such as sharing peripheral circuits across multiple arrays. Therefore, a systematic large-scale array evaluation remains essential for comparing these mapping schemes comprehensively and warrants future explorations.

Peripheral circuit precision compensation

Peripheral circuits introduce error sources including *IR* drop and AD/DA errors, which further degrade the MVM accuracy in analogue CIM. Mitigating such peripheral non-idealities to achieve precise results remains a key challenge and requires ongoing research.

The *IR* drop issue within memristor arrays is primarily from parasitic resistances, which include the inter-cell resistance and the access resistance¹⁵. This fundamentally restricts computation parallelism for analogue CIM. Proposed methods to mitigate this include applying column-specific scaling factors to restore distorted currents towards ideal values¹⁰¹, although their applicability in practical scenarios is limited. The 2T2R configuration⁷⁰ offers a distinct advantage by inherently reducing the accumulated source line current to mitigate *IR* drop while simultaneously enabling sign representation of synaptic weights, establishing it as a practical solution. Compact models that capture parasitic resistance-induced voltage drops have also been implemented, enabling pre-distortion compensation by modifying the target weight matrix (Fig. 4p). This approach has been shown to achieve accurate fully parallel matrix multiplication in a 128 kB array with all 1,024 inputs activated¹⁵.

DAC and ADC non-idealities, although often overlooked, also critically impact the computational accuracy. Research has indicated that they can degrade the model inference accuracy more severely than the device relaxation effect under certain conditions⁷⁷. To address this, a progressive parameter search scheme was proposed^{8,76}. It utilizes on-chip inference results and digital system simulations to find the optimal chip-operating conditions and store the offsets for subsequent compensations, as shown in Fig. 4q. A similar strategy has been proposed by leveraging results from on-chip and digital MVM computations, using a gradient descent approach to refine target weights¹⁰², as depicted in Fig. 4r. This iterative refinement progressively minimizes the MVM error, effectively mitigating both programming and peripheral circuit errors. By relying solely on the results of on-chip inference, this method circumvents the necessity for high-resolution ADCs to read individual devices, streamlining the overall system design. Whereas these approaches effectively mitigate array-level errors, the additional circuit overhead introduced by the necessary digital systems cannot be ignored.

Beyond the errors of peripheral circuits, the nonlinear *I*–*V* characteristics of memristor devices introduce another critical error source during analogue MVM operations as discussed earlier. Whereas strategies such as restricting the read voltage to a pseudo-linear subregion¹⁰³ or pre-tuning weights¹⁰⁴ have been explored, their effectiveness is often

limited by D2D variation. A more effective approach involves exploring novel array architectures and alternative signal domains¹⁷ (Fig. 4s). These include encoding inputs using fixed voltage pulses (for example, via pulse width or pulse number modulation) and encoding the computation results through non-current output schemes, such as the integration of output charge¹⁰⁵. Furthermore, some novel ADC designs^{106,107} and alternative output encoding schemes based on physical quantities such as pulse width¹⁰⁸ or delay time¹⁰⁹ are being co-developed with these architectures. Specifically, a voltage-regulating current-sense topology has been proposed to reduce the area and power consumption of conventional ADCs¹⁰⁶. In addition, a memristor-assisted adaptive ADC design has been developed, achieving increased energy efficiency and a smaller chip area¹⁰⁷. As for alternative output encoding schemes, MVM results can be translated into pulse width to enable efficient array-level cascading¹⁰⁸, or they can be encoded as the delay time of an inverter chain, enabling high computing parallelism in a static-power-free manner¹⁰⁹. Moreover, charge-domain CIM also offers an effective means to high energy efficiency and throughput¹¹⁰. These implementations show promising potential for achieving accurate signal readout with reduced overhead, collectively addressing the coupled challenges of peripheral non-idealities and device nonlinearity.

In addition, inspired by the error correction code (ECC) in digital memory, customized ECC for CIM systems has been explored¹¹¹ to enhance computational accuracy rather than ensure bit-level correctness (Fig. 4t)¹¹². For instance, Li et al. have demonstrated that on a DenseNet-40 network for CIFAR-10 classification, a tailored MAC-ECC (multiply-and-accumulate error correction code) can recover the inference accuracy from a 6.3% degradation to near-software baseline (only 1.1% loss)¹¹¹. Although promising for reducing inference errors and even achieving iso-accuracy in analogue CIM, this approach introduces area and power overheads from dedicated parity arrays and decoding circuits. Moreover, the efficacy of the ECC is bounded by the system's native bit error rate (or BER). Its advantage diminishes as the BER approaches the ECC scheme's capability and can fail completely beyond that threshold¹¹¹. More robust ECC schemes would improve coverage but at a cost of increasing circuit complexity and overhead, posing trade-offs between reliability and efficiency.

The overhead of these array-level precision compensation methods is qualitatively estimated in Supplementary Table 3. Although these concepts hold promise for reducing the error of analogue CIM systems, the associated overhead still demands careful design to balance performance and resource constraints.

Algorithm–architecture co-design

To implement state-of-the-art neural networks in practical applications, algorithm–architecture co-design becomes essential to achieve improved performance for memristor-based CIM. On the architecture side, it is favourable for the CIM hardware to integrate both analogue CIM and a digital core (such as digital CIM or a digital processor) as mixed-precision CIM (mix-CIM) to compensate for the accuracy limitations of analogue-only CIM systems, thus meeting diverse precision requirements of complicated algorithms. Meanwhile, on the algorithm side, optimizations are also needed to take the non-ideal characteristics of CIM hardware into consideration and make the algorithm error-tolerant, thereby enhancing the system's robustness. In this section, recent advances are reviewed on mix-CIM architectures and algorithm-level optimizations for error-tolerant CIM.

Mix-CIM architectures

Generally speaking, analogue CIM provides high energy efficiency, whereas digital cores ensure high computing accuracy. For complex neural networks, the flexible allocation of these hardware types, relying on the compiler and on-chip controller, is crucial to balance efficiency and accuracy. The compiler maps sensitive weights to digital cores via software simulation, and the controller manages data communication

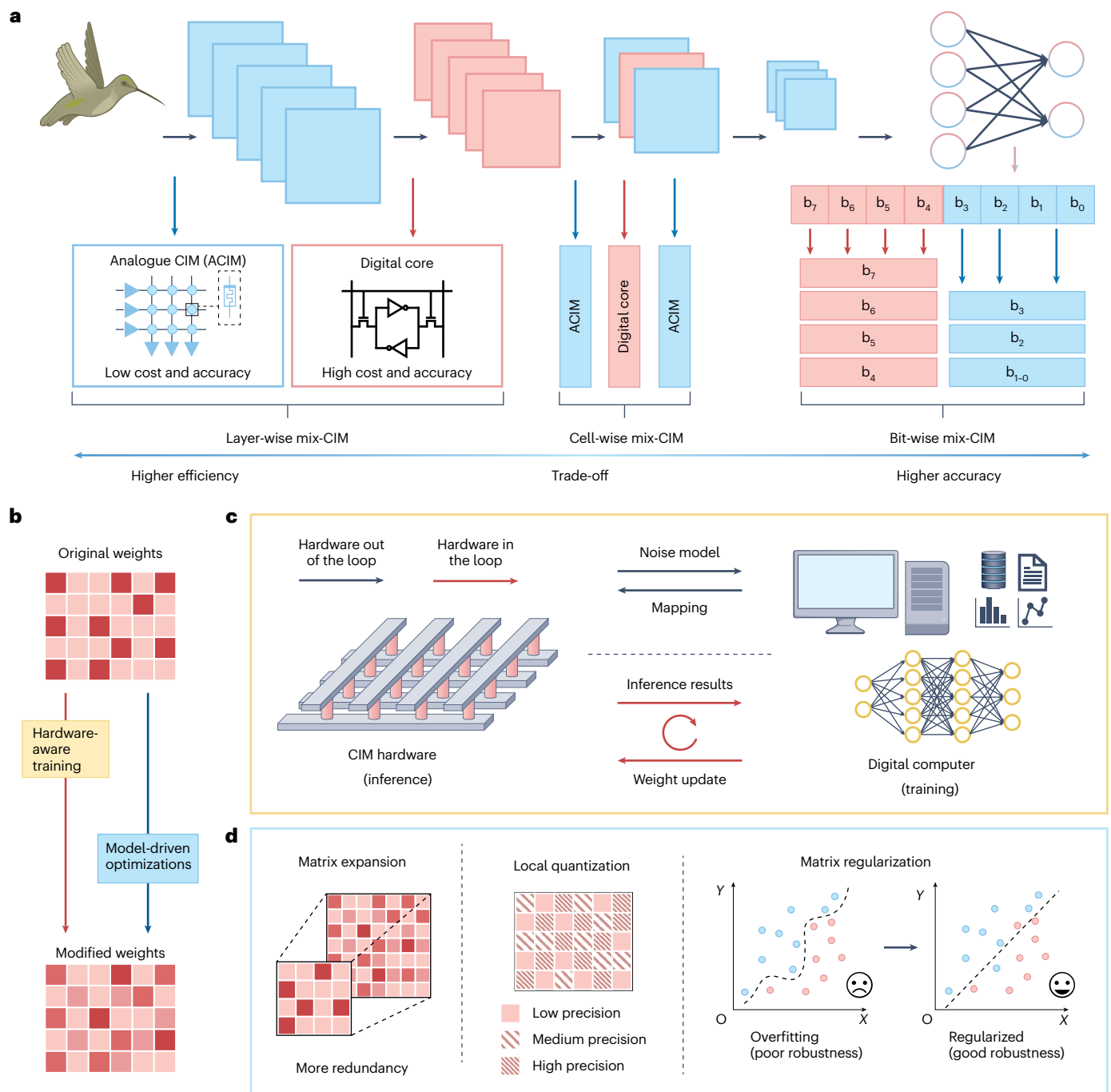


Fig. 5 | Algorithm-architecture co-design strategies. **a**, Illustration of mix-CIM architectures, including three types: layer-wise, cell-wise and bit-wise. **b–d**, Illustration of algorithm-level optimizations for error-tolerant CIM (**b**), which includes two major categories: hardware-aware training (**c**) and model-driven optimizations (**d**).

and synchronization between cores. Finer-grained allocation improves accuracy but increases compiler and controller overhead. Accordingly, current mix-CIM architectures are categorized by allocation granularity into layer-wise, cell-wise and bit-wise types (Fig. 5a).

Layer-wise mix-CIM. In DNNs, the input and output layers require higher accuracy, whereas hidden layers are more error-tolerant. Intuitively, the layer-wise mix-CIM architecture is proposed, in which different layers are mapped to different hardware domains to improve the overall computing accuracy. As a result, this strategy has been demonstrated to achieve <0.45% degradation on ResNet-20 and MobileNet-v2¹¹³. In terms of overhead, this coarse-grained layer-wise

approach imposes low stress on the compiler and incurs a minor communication cost between the analogue and digital cores.

Bit-wise mix-CIM. As illustrated in the rightmost panel of Fig. 5a, bit-wise mix-CIM offers the finest granularity of hardware allocation. In this approach, different bits (b_0 – b_7) of a single weight are mapped to different types of hardware. A typical strategy assigns most significant bits to digital cores for accuracy and least significant bits to analogue CIM for energy efficiency¹¹⁴. Analogue CIM can also operate in SLC (single-level cell) or MLC (multi-level cell) modes for finer trade-offs⁴⁶. Although it provides a promising performance, this bit-wise mix-CIM again places high burdens on the compiler and controller.

Cell-wise mix-CIM. Compared with the layer-wise and bit-wise approaches, the cell-wise strategy strikes a balance between computing accuracy and system overhead. It allocates hardware resources at the granularity of weight cells (for example, CNN (convolutional neural network) kernels, RNN (recurrent neural network)/LSTM gates or transformer attention heads), which is finer than the layer-wise approach but less complex than the bit-wise approach in compilation and data routing. Notably, this approach has been shown to achieve floating-point precision, enabling analogue CIM to handle regression tasks previously considered challenging¹¹⁵. Moreover, assigning sensitive or analogue-unsuitable fragments to the digital domain can improve the overall energy efficiency by enhancing hardware utilization.

In practical scenarios, strategies of different granularities can be jointly used for achieving a better performance, such as combining both the layer-wise and cell-wise mix-CIM¹¹³ or utilizing both bit-wise and layer-wise approaches⁴⁶. Moreover, the digital computation ratio must be precisely controlled, as excessive digital usage degrades the overall energy efficiency.

Algorithm-level optimizations for error-tolerant CIM

Owing to inherent hardware non-idealities in CIM, algorithm robustness is essential for reliability. Algorithm-level optimizations fall into two categories: hardware-aware training and model-driven approaches (Fig. 5b). Hardware-aware training retrains networks with hardware characteristics in mind, greatly improving the accuracy and scalability for complex networks, albeit with high preprocessing overhead. By contrast, model-driven optimizations rely on mathematical analysis without retraining, offering lower cost and better generalization, although typically yielding less accuracy gain and scalability.

Hardware-aware training. Hardware-aware training is further divided into two subcategories on the basis of hardware involvement during training (Fig. 5c). Hardware-out-of-the-loop (HOL) methods, such as noise-injected training¹¹⁶ and quantization-aware training¹¹⁷, incorporate a hardware non-ideality model into the training loop to enhance robustness. These approaches are easy to implement and can achieve accuracy that is comparable to floating-point implementations for large-scale DNNs⁷⁷. However, their effectiveness is often limited by discrepancies between the model and actual hardware. To address this, hardware-in-the-loop (HIL) methods integrate real CIM hardware into training by using measured array outputs for gradient calculation and weight updates. In situ learning, a common HIL approach, updates the network online after mapping to accommodate hardware non-idealities⁷, offering substantial accuracy gains but incurring high programming overhead due to repeated weight updates. To mitigate this, a chip-in-the-loop progressive fine-tuning strategy deploys the network layer by layer, retraining unmapped layers using actual outputs from mapped layers, achieving a 1.99% accuracy improvement on CIFAR-10 with ResNet-20⁸. Whereas HIL methods generally yield higher accuracy, they require tight collaboration between digital and CIM cores, leading to heavier preprocessing overhead.

Model-driven optimizations. Model-driven optimizations include three common approaches (Fig. 5d). Matrix expansion¹³ introduces redundancy by expanding a small weight matrix into a larger one, reducing the word error rate from 42 to 7.9% on LibriSpeech with MLPerf RNN¹³. However, it incurs additional hardware overhead due to larger array requirements and extra post-processing. Local quantization¹¹⁸ first calculates the weight sensitivity and then programs sensitive weights with higher precision while relaxing precision for less sensitive ones—improving the accuracy without increasing programming overhead. Matrix regularization addresses ill-posed problems that are highly sensitive to perturbations. Techniques such as singular

value truncation, Tikhonov regularization and ridge regression¹¹⁹ enhance robustness against hardware non-idealities¹²⁰. Although still in the early stages, this approach holds promise for broadening CIM applications.

For comparison, the overhead of these algorithm-level optimizations is roughly estimated and summarized in Supplementary Table 4. Generally speaking, hardware-aware training strategies, especially HIL approaches, tend to result in more overhead than model-driven optimization strategies. In practice, the above-mentioned methods may also be utilized together for better performance.

Outlook

Over the past 15 years, CIM hardware has evolved through three main stages (Fig. 6). In the first stage (~2010–2016), small-scale memristor arrays (≤ 1 kb) demonstrated MVM acceleration for simple tasks such as MNIST digit recognition, achieving an accuracy of 1–3 bits. The second stage (~2018–2020) saw larger arrays (1–100 kb) enabling more complex tasks such as image recognition and signal processing with 4–6-bit accuracy, claiming high energy efficiency. In the third stage (~2021–2025), CIM system-on-chip (SOC) technology emerged with ~100 kb to 100 Mb memristors, supporting more sophisticated workloads including regression tasks with 6–8-bit accuracy.

Looking ahead, CIM hardware is expected to continue advancing towards higher computing accuracy (>8 bits), larger integration density (>1 Gb) and more complex workloads (such as LLMs and artificial general intelligence (AGI)). To realize these goals, future CIM hardware should not only rely on continued optimizations at the individual device or array level but also pay more attention to a holistic, cost-aware co-design. Several important directions for future research are elaborated as follows (the rightmost panels of Fig. 6).

3D integration

Owing to the reticle size constraint and physical scaling limit of feature size, 3D integration technology becomes essential for high-density, high-performance CIM hardware. Different 3D integration technologies could be adopted, including device-level (for example, 3D vertical memristors), layer-level (for example, monolithic 3D integration) and chip-level (for example, chiplet) 3D integrations. These technologies could enable a much higher integration density (>100 Mb mm⁻²), better versatility (for example, mixed-precision computing cores, on-chip buffers and routers) and a higher interconnect bandwidth (>1 TB s⁻¹). Despite those promising advantages, 3D integration may introduce other critical challenges to the reliability and accuracy of analogue CIM that merit further studies to address them. For example, such increased fabrication process complexity may lead to yield loss and extra reliability issues, and the elevated temperature due to limited thermal dissipation in 3D stacked structures may degrade device conductance stability and impact the computing accuracy of analogue CIM. Besides, the increased coupling, parasitics and noise from 3D interconnects may introduce signal-integrity issues.

Analogue–digital mixed-CIM architecture

For more sophisticated scenarios such as LLMs and AGI, an adaptive CIM system is envisioned by integrating both analogue and digital cores and dynamically partitioning computations between these cores on the basis of specific precision demands, thus optimizing the trade-off between accuracy and energy efficiency. Meanwhile, this mixed-CIM architecture could greatly expand the applicability of analogue CIM, spanning from low-power edge devices to AI infrastructures and supercomputers.

End-to-end toolchain

The wide gap between memristor device models and overall CIM system performance makes both chip design and utilization inefficient

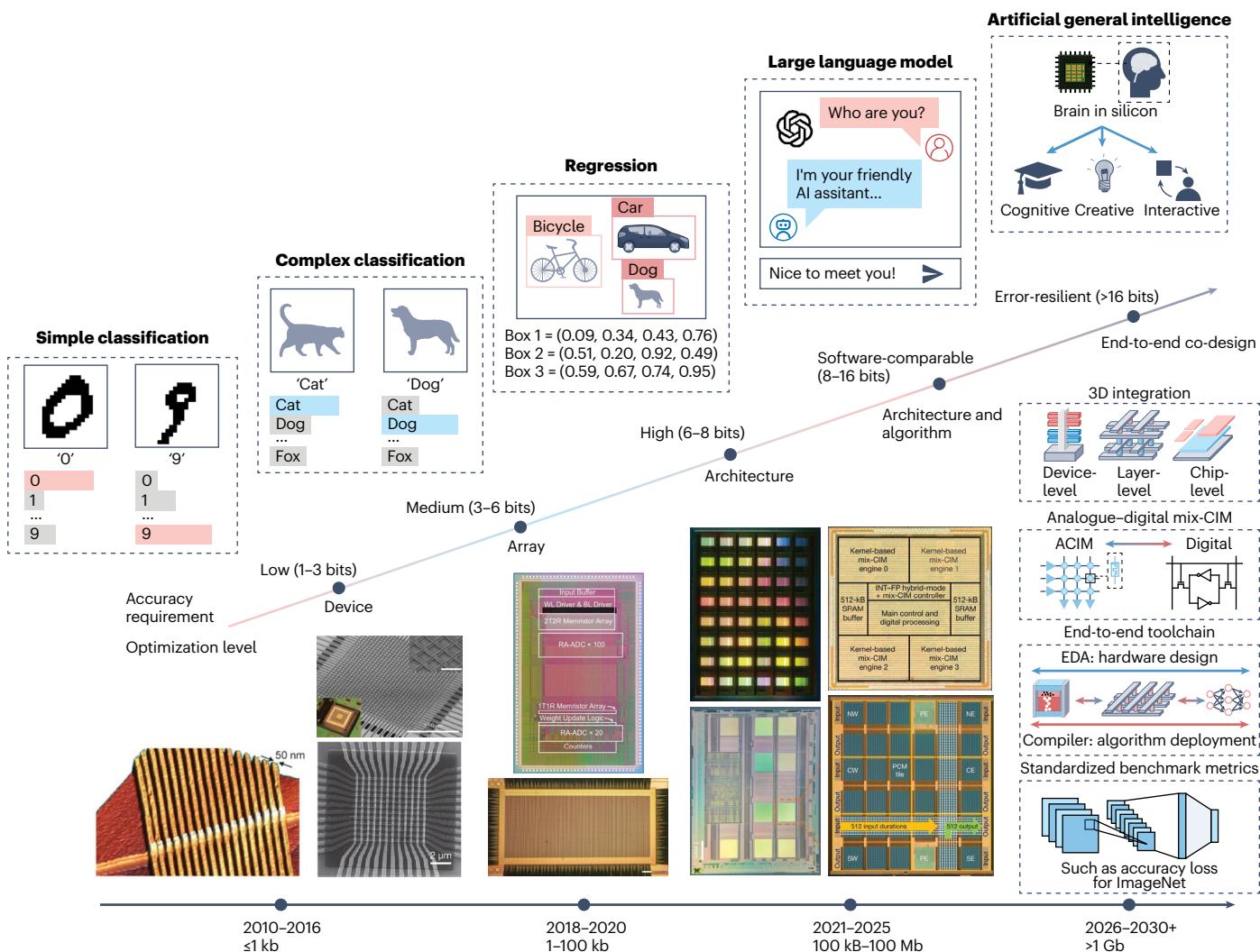


Fig. 6 | Roadmap for accurate and efficient analogue CIM. Schematic overview showing the progress in analogue CIM hardware over the past 15 years, with an outline of potential directions for future research. WL, word line; BL, bit line; INT, integer; FP, floating point; SRAM, static random access memory; NW, north west; NE, north east; CW, centre west; CE, centre east; SW, south west; SE, south east; PE, processing element; PCM, phase-change memory; EDA, electronic

design automation. Inset images from left to right and top to bottom reproduced with permission from: ref. 121, Springer Nature Limited; ref. 64, Springer Nature Limited; ref. 5, Springer Nature Limited; ref. 9, AAAS; ref. 68, Springer Nature Limited; ref. 8 under a Creative Commons license CC BY 4.0; ref. 11, Springer Nature Limited; ref. 113, Springer Nature Limited; ref. 13 under a Creative Commons license CC BY 4.0.

and time-consuming. To bridge this gap, a hardware-calibrated end-to-end design and simulation platform becomes critical. On the one hand, substantial efforts are needed towards developing a comprehensive electronic design automation toolchain for automated accuracy-overhead trade-off analysis, which would accelerate the chip design and improve the performance. On the other hand, an efficient CIM compiler is also essential for appropriate weight deployment and flexible dataflow scheduling. Together, this toolchain would greatly ease the development burden and foster a viable ecosystem for analogue CIM.

Standardized benchmarks and metrics

Previous demonstrations of analogue CIM typically rely on diverse datasets, making it almost impossible to conduct comprehensive performance analysis and fair comparisons. To address this, a standardized benchmarking framework is urgently needed. A comprehensive set of metrics should quantify not only the energy efficiency (TOPS W^{-1} , where TOPS denotes trillions of operations per second) or latency but also the accuracy degradation of analogue CIM. Therefore, industry-wide benchmarks are needed to define application-specific

accuracy thresholds (for example, a <1% accuracy loss compared with the software baseline for CIFAR-10 with the ResNet-20 model) while quantifying the acceptable overheads (area, latency, energy and so on) for computing error-mitigation techniques. In Supplementary Table 5, we propose a detailed benchmark table and encourage researchers to report detailed metrics of their optimized CIM systems, including the computing accuracy, programming overhead and inference overhead. Establishing such standardized benchmarks and metrics could enable transparent and quantitative comparisons across different optimization techniques. This, in turn, would accelerate the development of analogue CIM systems, facilitate reproducibility and guide future research towards commercially deployable CIM hardware.

In a nutshell, previous research has demonstrated that memristor-based analogue CIM offers great advantages in energy efficiency, outperforming traditional digital hardware by orders of magnitude. Moving forwards, through collaborative innovations across different hierarchical levels, analogue CIM could evolve from a promising concept into a viable and scalable technology, fully unlocking its potential to accelerate next-generation AI applications.

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Author contributions

J.T. and H.W. conceived of the idea and supervised the research. Z.J., H.Z. and J.T. wrote the paper. All authors provided input and reviewed the paper.

Competing interests

The authors declare no competing interests.

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